INCH-POUND
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PERFORMANCE SPECIFICATION

POWER SUPPLIES, PP-2953()/U AND PP-6224()/U

This specification is approved for use by the Communications- Electronics Life Cycle Management Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification establishes the performance and verification requirements for each of the following Power Supplies: the PP-2953()/U and the PP-6224()/U . The power supply furnishes 0-25 amperes over a voltage range of 24 to 32VDC from nominal 115/230 VAC at 50, 60 or 400 Hz single phase. The power supply can also operate from a 24 VDC storage battery in the battery standby mode.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, US Army US Army Communications-Electronics RDEC, ATTN: RDER-PRQ-QE, Fort Monmouth, NJ 07703-5201 or emailed to Robert.Francis@us.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595/34094	-	Colors Used in Government Procurement

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-53072	-	Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection
MIL-DTL-55181	-	Connectors, Plug and Receptacle, Intermediate Power (Electrical, Waterproof), Type MW, General Specification For
MIL-DTL-64159	-	Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-3009	-	Lighting, Aircraft, Night Vision Imaging System(NVIS) Compatible
MIL-STD-13231	-	Marking of Electronic Items

(Copies of these documents are available online at https://assist.daps.dla.mil/online/ from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

A3286985	-	PP-2953D/U & PP-6224C/U Housing Assembly/Mounting Requirements
SM-D-667917	-	Cable Assembly, Power, Electrical CX-11979/U
SM-B-667921	-	Connector, Receptacle
SM-D-667953	-	Cable Assembly, Power, Electrical CX-12342/U

(Copies of these drawings are available from Commander, US Army CECOM Life Cycle Management Command, ATTN: AMSEL-LCL-ECM, Fort Monmouth, NJ 07703)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC 61000-4-2	-	Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test
IEC 61000-4-5	-	Electromagnetic compatibility (EMC) – Part 4-5: Testing and measurement techniques - Surge immunity test

(Copies of these specifications are available online from International Electrotechnical Commission http://webstore.iec.ch)

UNDERWRITERS LABORATORIES

UL 60950-1	-	Information Technology Equipment

(Applications for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096, or at http://ulstandardsinfonet.ul.com)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subject to first article inspection in accordance with 4.2.

3.2 Parts and materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein.

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3.3 Design/construction. The power supply shall be designed to obtain minimum size and weight with maximum reliability consistent with the overall requirements. Design techniques shall be used that provide stability under static and dynamic load conditions and minimize Electromagnetic Interference (EMI). The functional requirements of the Power Supply shall be as specified in this specification.

3.3.1 System configuration. The power supplies shall include the following cable assemblies as part of the system configuration:

Power Supply	AC input cable (see 3.3.3.1)	DC output cable (see 3.3.3.2)	Remote sensing cable connector (see 3.3.3.3)
PP-6224()/U	1 per system	1 per system	1 per system
PP-2953()/U	1 per system	2 per system	-

3.3.2 Dimensions. The envelope dimensions (chassis) of the Power Supply shall conform to Drawing A3286985 to ensure the mounting requirements (for bench-mounted capability) and the form and fit of the device. The physical dimensions of the Power Supply shall not exceed the following:

Height 7.0 inches
Width 14.25 inches
Depth 14.50 inches

3.3.2.1 Cover. A cover shall be installed to protect the front panel from damage during transit.

3.3.3 Cables.

3.3.3.1 AC cable. The AC Input Cable shall be capable of mating with the AC input connector (see 3.3.4.2) and shall be the CX-11979()/U 15FT. The CX-11979()/U 15FT shall conform to the dimensional requirements of Drawing SM-D-667917 to ensure proper interface requirements and shall be 15 feet in length. Contacts A and B shall be line, contacts C and D shall be neutral, and contact E shall be ground. This drawing is referenced for dimensional and interface purposes only.

3.3.3.2 DC cable. The DC Output Cable shall be capable of mating with the DC input connector (see 3.3.4.1) and shall be the CX-12342( )/U 3FT. The CX-11979( )/U 3FT shall conform to the dimensional requirements of Drawing SM-D-667953 to ensure proper interface requirements and shall be 3 feet in length. This drawing is referenced for dimensional and interface purposes only.

3.3.3.3 Remote sensing cable connector. The PP-6224()/U shall include a connector capable of mating with the DC output connector (front panel) with remote sensing capability (see 3.3.4.4). The connector shall be suitable for use in fabrication at the field level of a remote sensing cable.

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3.3.4 Connectors.

3.3.4.1 DC input connector. A connector, part number M55181/2-01 or equivalent, shall be provided for DC input in the battery standby mode (see 3.7.2). Contact A shall be negative (-), contact B shall be positive (+), and contacts C and D are not used. This connector shall be located on the rear panel of the power supply.

3.3.4.2 AC input connector. One AC input plug connector shall be provided on the front panel. The AC input connector (Part Number PT07C14-5P or equivalent) shall be capable of mating with the AC Cable, CX-11979( )/U.

3.3.4.3 DC output connector. Two DC output connectors shall be provided for DC output power, part number M55181/4-01 or equivalent. Contact A shall be negative (-), contact B shall be positive (+), and contacts D and C are not used. Both connectors shall be located on the rear panel of the power supply.

3.3.4.4 DC output connector (front panel) with remote sensing capability. One DC output connector shall be provided on the front panel with remote sensing capability. This connector shall conform to the dimensional requirements of Drawing SM-B-667921 (Part No. PT07C-14-16S). This drawing is referenced for dimensional and interface purposes only. Contact A shall be positive (+) sense, contact B shall be negative (-) output, contact C shall be negative (-) sense, and contact D shall be positive (+) output.

3.3.5 Indicator lights/output meters. All indicator lights and output meters used shall be Class B Night Vision Imaging System (NVIS) (see 6.5.6) compatible in accordance with the requirements of MIL- STD-3009 for “warnings, caution, and advisory displays and indicators” used on ground vehicles and equipment (see paragraphs 3.1.4, 3.2.2 (secure), 3.2.3 (friendly), 3.2.4 (covert), 3.6, and Appendix C of MIL-STD-3009). NVIS compatibility assures the operation of the power supply without degrading the image intensification capabilities of the NVIS during night operations. Indicator lights and output meters used shall also maintain daylight readability and legibility. This requirement shall be certified.

3.3.5.1 DC emergency indicator light. The DC Emergency Indicator Light shall indicate that the power supply is operating in the Battery Standby Mode (see 3.7.2).

3.3.5.2 Normal indicator light. The Normal Indicator Light shall indicate the power supply is producing regulated dc output using ac input.

3.3.5.3 AC indicator light. The AC Indicator Light shall indicate that ac input is being provided to the power supply.

3.3.5.4 Output meter. The Output Meter shall be capable of displaying dc output voltage or current.

3.3.5.4.1 Output meter switch. The Output Meter switch shall provide the capability to select either the output voltage or output current to be displayed on the output meter.

3.3.6 DC adjustment. The DC Adjustment shall allow for desired dc output voltage.

3.4 Weight. The total weight of the Power Supply shall not exceed 36 pounds (excluding the cables). The total weight of the Power Supply in its packaging and shipping requirements shall not exceed 44 pounds.

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3.5 Repair authorization. Except upon prior specific approval of the contracting officer, printed wiring assemblies shall not be repaired.

3.6 Marking of printed circuit boards. All printed circuit boards shall be marked per MIL-STD-13231 with the following information: Manufacturer's part number, CAGE CODE, and NSN (if applicable).

3.7 Performance characteristics. The Power Supply shall be capable of providing power for up to two loads and shall meet the following electrical requirements:

3.7.1 Input voltage (AC). The power supply shall deliver the output required by this specification (see 3.7.3) from the following single phase input sources without any user interaction required:

Volts ac		Frequency (Hz)	
100-253	-	50,60	(±5%)
115	(±10%)	400	(±5%)
230	(±5%)	400	(±5%)

This mode of operation shall be indicated an indicator light on the front panel labeled "DC NORM".

3.7.2 Power Changeover. The power supply shall maintain stable output voltage when the AC input drops below 100 VAC. The voltage shall either be the set voltage (see 3.3.6) or the voltage present at the DC input connector (see 3.3.4.1). The output voltage shall not rapidly oscillate between these two voltages. When operating on DC input power, it shall be indicated by an indicator light on the front panel labeled "DC EMER" and extinguishing of the "DC NORM" indicator light. There shall be consistency of polarity of the circuit connecting the "battery in" plug and the "power out" plug. (see 4.5.5).

3.7.3 Output voltage. The voltage across the output of the power supply shall be 28.0 Volts (±1%) over the load range of 0-25 amperes. The output voltage shall also be adjustable over the range of 24.0 to 32.0 volts by an external locking type control and shall be within ±1% of the voltage setting over the load range from 0 to 25 amperes (see 4.5.1).

3.7.4 Ripple voltage. The RMS ripple voltage shall not exceed 0.5 percent at any output condition from no-load to full-load. The peak to peak ripple voltage including switching spikes and other noise components shall not exceed 2% of the output, at any output condition from 1A to full-load. For loads under 1A, the peak to peak ripple voltage shall not exceed 5% of the output. The ripple voltage includes voltage perturbations within the frequency range of 20 Hz to 10 MHz. (see 4.5.2)

3.7.5 Recovery time. The output voltage of the power supply shall recover to 28.0 Volts ±1% within 25 milliseconds when the load is switched from no-load to full-load and from full-load to no-load. The maximum instantaneous overshoot voltage under these conditions and for any turn on or turn off condition of the power supply shall not exceed 20% of the nominal voltage (28.0 VDC). (see 4.5.3)

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3.7.6 Protection.

3.7.6.1 Over voltage. The power supply shall show no deleterious effects after being subjected to overvoltage input as tested. The power supply shall automatically resume normal operation when the fault condition is removed. Before and after testing as specified in 4.5.10, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.7.6.2 Overload protection. The power supply shall be protected from physical and electrical damage from any overload conditions (including a direct short circuit) by the use of re-settable circuit breakers in the "AC input" circuit and the "DC output" circuit, and by automatic current limiting circuitry. The power supply shall automatically resume normal operation when the fault condition is removed. Before and after testing as specified in 4.5.4, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.7.6.3 Current limiting. Automatic control circuitry shall limit the output current to a maximum of 30 amperes to protect the power supply from overload and automatically restore the output when the overload is removed. The power supply shall automatically resume normal operation when the fault condition is removed. Before and after testing as specified in 4.5.4, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.7.6.4 Electrostatic discharge (ESD). The power supply shall withstand ESD to ±8kV, contact discharge on any surface, panel component or connector, including exposed pins, in both operating and non-operating mode. Before and after testing as specified in 4.5.11, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2
Current limiting	3.7.6.3	4.5.4

3.7.6.5 Surge immunity. The power supply shall withstand ±2kV surge, line-to-line and line-to-ground on the AC input. Before and after testing as specified in 4.5.12, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

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3.7.7 Efficiency. The overall efficiency of the power supply shall not be less than 85% when tested as specified in 4.5.7.

3.7.8 Remote sensing. The power supply shall include provisions for extending the sensing of the DC output voltage through signal leads from the auxiliary output receptacle to the remote load terminals. Remote voltage, as tested in 4.5.6, need not exceed 29 VDC. Also, provisions shall be included to automatically revert to local sensing if the remote circuit is inadvertently opened. (see 4.5.6)

3.7.9 Stability. Output voltage drift at a constant load and ambient temperature shall not exceed 0.2% at any time over a period over a period of two (2) hours or 0.5% at any time over a period of 24 hours (see 4.5.8).

3.7.10 Temperature coefficient. Change in output voltage as a function of ambient temperature shall not exceed 0.02 percent per degree F (0.036 percent per degree C) over the entire specified operating temperature range. When tested as specified 4.5.9, the change in output voltage shall not exceed 1.00%.

3.8 Environmental conditions. The power supply shall meet the following environmental conditions:

3.8.1 Temperature. The power supply shall operate at ambient temperature in the range of -40°F (-40°C) to +150°F (+66°C). (The 150°F (66°C) temperature includes effect of solar loading.) The power supply (non-operating) shall withstand exposure in the range of -80°F(-62.2°C) to +160°F (+71°C).

3.8.1.1 Low. Before, during, and after testing as specified in 4.6.1.1, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2
Recovery Time	3.7.5	4.5.3
Power Changeover	3.7.2	4.5.5

3.8.1.2 High. Before, during, and after testing as specified in 4.6.1.2, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2
Recovery Time	3.7.5	4.5.3
Power Changeover	3.7.2	4.5.5

3.8.2 Humidity. The power supply shall withstand elevated temperature-humidity conditions for not less than ten (10) days. When tested as specified in 4.6.2, the power supply shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet full specification performance during the specified times for the following measurements:

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Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.8.3 Low pressure (altitude). The power supply shall operate up to 15,000 feet above sea level and shall withstand up to 40,000 feet above sea level. Before, during, and after testing as specified in 4.6.3, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.8.4 Immersion. Before and after testing as specified in 4.6.4, the power supply shall show no evidence of leakage and meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.8.5 Vibration and ballistic shock. The power supply shall experience no deleterious effects from vibration (see 4.6.5), bench handling (see 4.6.6), or ballistic shock (see 4.6.7) when tested. Full specification performance for the following measurements shall be met before and after testing:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.8.6 Fungus. The power supply, when tested, shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.5.5) as evidenced by growth colonization (see 6.5.3) which includes branching (see 6.5.1) and sporulation (see 6.5.7) on or within each cubic foot, or fraction thereof, of power supply assembly volume. Isolated instances of partial tubular germination (see 6.5.8) shall not be included in this evaluation.

3.8.7 Sand and dust. The power supply shall operate in its normal operational configuration, while exposed to blowing dust and sand environments as tested in 4.6.9. Failure of the power supply to maintain the required output during testing shall constitute a failure of this test. Before and after testing as specified, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

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3.8.8 Rain. The power supply shall operate in its normal operational configuration while exposed to rain. Failure of the power supply to maintain the required output during testing shall constitute a failure of this test. Before and after testing as specified in 4.6.10, the power supply shall meet full specification performance for the following measurements:

Measurement	Reqt Para.	Test Para.
Input and Output Voltage	3.7.1, 3.7.3	4.5.1
Ripple Voltage	3.7.4	4.5.2

3.9 Electromagnetic interference (EMI). The power supply shall comply with the applicable emissions and susceptibility requirements of MIL-STD-461 Table V, Aircraft, Army, Including Flight Line (see 4.8). Susceptibility tests shall be monitored for conformance with the ripple voltage requirements of 3.7.4 of this document. RS103 shall be performed from 10 KHz to 400 MHz as follows:

Frequency Range	Susceptibility Level
10 KHz to 1.99 MHz	1 Volt/meter
2.0 MHz to 29.99 MHz	5 Volt/meter
30 MHz to 400 MHz	10 Volt/meter

3.10 Air-seal. The power supply shall be capable of maintaining a negative pressure. The decrease in vacuum shall not exceed 0.1 pound per minute when tested. The case shall provide means for enabling performance of the air-seal test (see 4.6.11 and 6.4).

3.11 Reliability. The power supply shall have a 90% probability of providing 500 hours of failure-free operation under the conditions listed in a) through c) below. When tested in accordance with 4.10, no chargeable failures shall occur during the duration of the required test time (see 6.5.2 for chargeable failure definition). Test items shall meet the requirements of 3.7.3 and 3.12.

a) Ambient temperature of 122°F (50°C).
b) Duty cycle-intermittent operation.
c) Environment-ground mobile.

The above requirement shall be verified in accordance with 4.10.

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3.12 Workmanship. Workmanship shall be in accordance with 3.12.1 for the following:

a) general requirements for plastic material and plastic parts
b) mechanical operation of controls
c) printed wiring assemblies and printed wiring boards
d) screws & other threaded devices and related parts
e) cleaning
f) riveting
g) soldering
h) welding
i) internal wiring practices including: slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable
j) marking
k) finish, protective
l) CARC

3.12.1 Workmanship requirements. This requirement establishes the acceptance workmanship criteria for electronic equipment.

3.12.1.1 Cleaning. After fabrication, parts and assembled equipment shall be cleaned of smudges; loose, spattered or excess solder; weld metal; metal chips and mold release agents; or any other foreign material which might detract from the intended operation, function or appearance of the power supply.

3.12.1.2 Threaded part or devices. Screws, nuts and bolts shall show no evidence of cross threading, mutilation, or detrimental or hazardous burrs, and shall be firmly secured.

3.12.1.3 Wiring. Wires and cables shall be positioned or protected to avoid contact with rough or irregular surfaces and sharp edges and to avoid damage to conductors or adjacent parts.

3.12.1.4 Shielding. Shielding on wires and cables shall be secured in a manner that will prevent it from contacting or shorting exposed current-carrying parts. The ends of the shielding or braid shall be secured to prevent fraying.

3.12.1.5 Exterior surfaces. All exterior surfaces shall be lusterless.

3.13 Requirements for applying Chemical Agent Resistant Coating (CARC).

3.13.1.1 Technical requirements. Exterior surfaces of the Power Supply shall be painted with Lusterless Green 383 CARC in accordance with MIL-DTL-64159 or MIL-DTL-53039 (FED-STD-595 Color No. 34094). All coatings shall be obtained from a Qualified Products List (QPL) source. Materials and processes used in the application of all preconditioning, passivation, pretreatments, primers and final films shall be in accordance with MIL-DTL-53072 and shall meet all Verification Provisions of MIL-DTL-53072.

3.13.2 Visual inspection criteria (verification). Color match on the finished assembly shall be per paragraph 3.3.3 of MIL-DTL-64159 or MIL-DTL-53039, as applicable.

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3.13.3 Touch up. When piece parts have been finished with complete paint finishes, fixed assemblies shall be touched up in order to provide a continuous film across the interfaces. This requirement does not apply to hairline joints between matching parts, gasketed joints, or where moving parts such as hinges or catches are present. Touch-up paint used shall be identical to the final paint film and shall provide a reasonable color match with that of other assembled parts.

3.13.4 Coating thickness. The dry film thickness of each pretreatment, primer and final film coating shall be in accordance with MIL-DTL-53072.

3.13.5 Exceptions. Parts which can be flexed should not be painted with CARC since the paint will crack.

3.13.6 Marking for CARC. The power supply shall be marked with 'CARC' stenciled in CARC paint in close proximity to the nameplate. A contrasting color shall be used for the marking (e.g. black marking on a green surface). The marking shall be durable enough to withstand the full environmental conditions the surface will be exposed to during field use. Plates or embossed lettering may be used. Embossed letters need not be highlighted as long as they are legible

3.14 Marking. Marking shall conform to MIL-STD-13231. Front panel marking shall be Group I as described in MIL-STD-13231.

3.14.1 Identification label. An identification label shall be provided for each power supply. The material used for this label shall be non-reflective. The label shall have a light background with contrasting black text. For the PP-6224()/U, the upper right hand corner of the label shall be shaded black and shall contain the nomenclature in contrasting text as in Figure 1.

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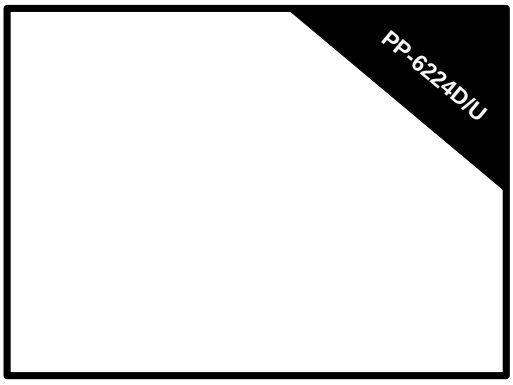


Figure 1- Representative PP-6224()/U label

The label shall include the following information:

a) Power supply nomenclature
b) NSN
c) Contract number
d) Serial number
e) Manufacturer
f) Manufacturer's part number
g) Manufacturer's CAGE CODE

The label shall be located on a visible side of the power supply. The material used for the label shall be weather-resistant and capable of withstanding the tactical and environmental requirements of this specification.

3.15 Visibility. Whenever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with minimum disassembly of the power supply.

3.16 System safety.

3.16.1 General safety. The power supply shall be Listed to UL 60950-1 and bear the Listing mark. In lieu of Listing, the power supply shall be tested per 4.9.1, 4.9.2, and 4.9.3 and meet the visual and mechanical requirements of UL 60950-1.

3.16.2 Ground continuity. When tested as specified in 4.9.4, the resistance to the grounding pin shall not be greater 0.1 ohms.

3.16.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

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4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a) First article inspection (see 4.2),
b) Inspection covered by subsidiary documents (see 4.3),
c) Conformance inspection (see 4.4).

4.2 First Article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.2.1 First article units. Unless otherwise specified, the contractor shall furnish ten (10) first article units of the complete power supplies. The presence of one or more defects shall be cause for rejection of First Article.

4.2.2 First article inspection. The first article inspection shall consist of inspections specified in TABLE I. Tests shall be performed in the order indicated for each test sample.

4.3 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of power supply before preparation for delivery:

Item	Requirement Paragraph.
CARC	3.13
Marking	3.14

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TABLE I. First Article Inspection.

Inspection	Reqt Para	Test Para	UNIT									
			1	2	3	4	5	6	7	8	9	10
Inspections covered by subsidiary documents		4.3	Inspections to be performed on all units.									
Group A Inspection	See TABLE II		Inspections to be performed on all units.									
Group B Inspection	See TABLE IV		Inspections to be performed on all units.									
Frequency-voltage-load matrix	3.7.3	4.5.1.2	Inspections to be performed on all units.									
Temperature Test	3.8.1	4.6.1	X				X					
Relative Humidity	3.8.2	4.6.2	X									
Temperature Coefficient	3.7.10	4.5.9	X				X					
Altitude (Elevation)	3.8.3	4.6.3		X								
Stability: Method 2	3.7.9	4.5.8		X								
Electromagnetic Interference	3.8.8	4.8						X				
Vibration	3.8.5	4.6.5			X	X						
Bench handling	3.8.5	4.6.6			X	X						
Shock, Ballistic	3.8.5	4.6.7			X	X						
Immersion	3.8.4	4.6.4			X							
Sand and dust	3.8.7	4.6.9			X							
Rain	3.8.8	4.6.10				X						
Fungus*	3.8.6	4.6.8							X			
ESD	3.7.6.4	4.5.11								X		
Surge immunity	3.7.6.5	4.5.12								X		
Touch current**	3.16.1	4.9.1								X		
Reliability	3.11	4.10									X	X

*Note: The FAT sample undergoing Fungus inspection shall be disposed of properly.
**Note: Testing not required if product is Listed by Underwriter Laboratories

4.4 Conformance inspection. Inspection of products for delivery shall consist of Groups A, B, and Group C when specified.

4.4.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections in TABLE II. Power Supplies which have passed Group A inspection shall be formed into lots.

4.4.1.1 Group A nonconformance. Units which do not pass Group A inspections may be reworked. The entire Group A inspection for the failed unit must be repeated after rework. Printed wiring assemblies shall not be repaired (3.5).

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4.4.1.2 Order of inspection within Group A. Group A inspection shall be performed in any order, provided Stability is performed first.

TABLE II. Group A inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Stability: Method 1	3.7.9	4.5.8
Input voltage (AC)	3.7.1	4.5.1
Output voltage	3.7.3	4.5.1
Power changeover	3.7.2	4.5.5
Ripple voltage	3.7.4	4.5.2
Efficiency	3.7.7	4.5.7
Over voltage	3.7.6.1	4.5.10
Current limiting	3.7.6.3	4.5.4
System safety	3.16	4.7.1
Grounding Continuity	3.16.2	4.9.4
Air-seal	3.10	4.6.11
Visual and mechanical	3.12	4.7

4.4.2 Group B inspections. Group B inspections shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. Sampling of lots shall be in accordance with TABLE III below. No failures shall be allowed. Tests shall be conducted in the order shown in TABLE IV in accordance with the inspections listed in TABLE IV.

4.4.2.1 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE III. Group B sampling.

Lot Size	Samples Required 1/
Up to 13	ALL
14 to 500	13
501 to 1200	20
Over 1200	32

NOTES:
1/ No failures allowed.

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TABLE IV. Group B inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Recovery time	3.7.5	4.5.3
Overload protection	3.7.6.2	4.5.4
Remote sensing	3.7.8	4.5.6
Bench handling	3.8.5	4.6.6
Electric Strength**	3.16.1	4.9.2
Retained Charge**	3.16.1	4.9.3

**Note: Testing not required if product is Listed by Underwriter Laboratories

4.4.3 Group C inspection. Group C inspection shall consist of the inspections identified in TABLE V below. Samples shall be selected in accordance with 4.4.3.1.1.

TABLE V. Group C inspection.

Inspection	Requirement Paragraph	Inspection Paragraph	Sample Numbers
Frequency - voltage - load matrix	3.7.1	4.5.1.2	1-4
Stability: Method 2	3.7.9	4.5.8	1-4
Temperature coefficient	3.7.10	4.5.9	1
Temperature test	3.8.1	4.6.1	2
Immersion	3.8.4	4.6.4	2
Vibration	3.8.5	4.6.5	3
Shock, ballistic	3.8.5	4.6.7	4

4.4.3.1 Group C sampling.

4.4.3.1.1 Sampling for inspection (Group C). Four samples shall be selected at random from units having passed Group B inspections within12 months after First Article completion and within 12 months of each prior Group C sampling,. The samples shall be numbered 1 through 4. These units shall have passed Group A and B inspections. Each sample shall be subjected to each inspection specified for the specific sample number.

4.4.3.2 Group C failures. Actions relative to Group C failures shall be as specified in the contract.

4.4.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are re-subjected to and pass Group A inspection and the frequency-voltage-load matrix inspection of 4.5.1.2 (see Sub-group I, TABLE V). Printed wiring assemblies shall not be repaired (3.5).

4.5 Electrical tests.

4.5.1 Voltage tests, input and output. Whenever test requirements call out this paragraph, both sub-paragraphs apply.

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4.5.1.1 Input and Output. The output voltage shall meet the requirements of 3.7.3 over the load current range of 0 to 25 amperes and over the input voltage range 100 to 126.5 volts/ 207 to 253 volts, 60 Hz.

4.5.1.2 Frequency-voltage-load matrix test. The output voltage shall meet the requirements of 3.7.3 when the load, input voltage (using appropriate input connector), and frequency are varied as specified below; also when the external control is adjusted over the output voltage range:

Input Voltage	Type	Frequency	Load (amps)
100	VAC	47.5, 63, 420	0 to 25
115	VAC	47.5, 63, 420	0 to 25
126.5	VAC	47.5, 63, 420	0 to 25
207	VAC	47.5, 63, 420	0 to 25
230	VAC	47.5, 63, 420	0 to 25
241.5	VAC	420	0 to 25
253	VAC	47.5, 63	0 to 25

4.5.2 Ripple voltage. The peak to peak ripple voltage shall be measured with suitable equipment having a bandwidth of 20 Hz to 10 MHz . This test shall be made using the following input voltages: 103.5 VAC- 60 Hz, 115 VAC- 60 Hz, and 126.5 VAC – 60 Hz. (see 3.7.4).

4.5.3 Recovery time. With the power supply operating from 115 VAC – 60 Hz, the load shall be switched from 25 to 0 amperes and the power supply switched on and off at each of the load conditions. The voltage transients shall be noted on suitable equipment. This procedure shall be repeated when the load is switched from 0 to 25 amperes. The transient voltage shall meet the requirements of 3.7.5.

4.5.4 Overload protection and current limiting. With the power supply operating at full load (25 amps) the load shall be increased gradually to a short circuit condition and the output current noted during the current limiting condition. Upon removal of the short, the power supply shall automatically return to normal operating conditions. Repeat five (5) times. The power supply shall then meet the requirements of 3.7.1 , 3.7.3, and 3.7.4. This test shall be performed at room temperature and during the temperature tests of 4.6.1. The input voltage during the tests shall be 115 V RMS, 60 Hz. Visual examination shall be made for any physical damage that might have occurred due to the overload conditions. (see 3.7.6.2 and 3.7.6.3). At the end of the test, the power supply shall be tester per 4.5.1 and 4.5.2 and meet the requirements of 3.7.1, 3.7.3, 3.7.4.

4.5.5 Power changeover. Connect a 24V nominal DC source (battery or power supply) to the DC input connector (see 3.3.4.1). Using a 115 V, 60 Hz single phase variable supply, reduce the AC input voltage until the power supply automatically changes over to DC and note the AC voltage at this point. Increase the AC input voltage until the power supply automatically changes back to AC and note the value of AC input voltages at this point. Repeat this test using 115 VAC – 400 Hz. This test shall be conducted with both a 2 amp and a 25 amp load for each input voltage and frequency specified. Proper “light” indication shall be checked for conformance to requirements of 3.7.2. Under the power changeover condition, indicated when the indicator light on the front panel labeled “DC EMER” is on, a check shall be performed to determine the continuity and consistency of polarity of the circuit connecting the “battery in” plug and the “power out” plug.

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4.5.6 Remote sensing. With the power supply adjusted for remote sensing, the load shall be connected through a 25 foot cable, 4 conductor #12 wire size (2 conductors for sensing). While in this configuration, the output voltage shall be measured at the load and shall meet the output voltage requirements of 3.7.3 when tested as specified in 4.5.1. During this test there shall be no evidence of unstable operation (see 3.7.8).

4.5.7 Efficiency. The overall efficiency of the power supply shall be measured under conditions of full load, 28.0 volts at 25 amperes, and at an input of 115 volts RMS, and at frequencies of 50, 60 and 400 Hz. The percent efficiency shall be determined by the following and shall meet the requirements of 3.6.6:

$$\frac{Power_{out}}{Power_{in}} \times 100 = efficiency$$

4.5.8 Stability. The power supply shall be operated from 115 volt, 60 Hz source with a 25 ampere load at nominal output voltage at room temperature.

Method 1: The output shall be monitored continuously for not less than 2 hours. The power supply shall meet the requirements of 3.7.9.

Method 2: The output shall be monitored continuously for not less than 24 hours. Repeat this test using an input of 115 VAC, 400 Hz. The power supply shall meet the requirements of 3.7.9.

4.5.9 Temperature coefficient. The power supply shall operate from 115 volt, 60 Hz source with a 25 ampere load at nominal output for 4 hours in an ambient temperature of 70°F (21.1°C). The ambient temperature shall then be increased to 120°F (49°C) and the power supply operated for 4 hours. Repeat this test at an ambient temperature starting at -40°F (-40°C) and increasing to +10°F (-12.2°C). The power supply shall meet the requirements of 3.7.10.

4.5.10 Over voltage. With the AC input adjusted to 115 VAC 60 Hz, adjust the DC output to 28 VDC with a 25 ampere load. In no greater than two (2) seconds, raise the input voltage to 139 VAC. Maintain this voltage for no less than one (1) minute and then reduce it back to 115 VAC. Repeat the test using 230 VAC, 60 HZ, raised to 278 VAC. At the end of the test, the power supply shall be tested per 4.5.1 and 4.5.2 and meet the requirements of 3.7.1, 3.7.3, 3.7.4 and 3.7.6.1.

4.5.11 Electrostatic discharge protection. The power supply shall continue to operate properly after subjection to IEC 61000-4-2, Level 4, to include direct contact discharge to the contacts of each DC output connector. The unit shall be tested operating and non-operating with all switches in the on position. Operation shall be deemed proper by applying a known load to the power supply before and after the test. The current shall be within 5% of its initial value after the test. At the end of the test, the power supply shall conform to the requirements of 3.7.3, 3.7.4, and 3.7.6.3. At the end of the test, the power supply shall be tester per 4.5.1, 4.5.2, 4.5.4 and meet the requirements of 3.7.1, 3.7.3, 3.7.4, and 3.7.6.3 (see 3.7.6.4) .

4.5.12 Surge immunity. The power supply shall continue to operate properly after subjection to IEC 61000-4-5, Class 4, Level 3, 230VAC, 50Hz or 60Hz AC input. Surge voltage applied during the test need not exceed 2kV. The unit shall be tested operating with all switches in the on position. At the end of the test, the power supply shall be tester per 4.5.1 and 4.5.2, and meet the requirements of 3.7.1, 3.7.3, and 3.7.4 (see 3.7.6.5).

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4.6 Environmental conditions.

4.6.1 Temperature test.

4.6.1.1 Low temperature. The power supply shall be tested in accordance with Method 502, Procedure I of MIL-STD-810. The storage temperature shall be -80°F (-62.2°C). Upon completion of Step 4, adjust the internal chamber temperature to -40°F (-40°C) and maintain until temperature stabilization of the test item is reached. Configure the power supply in its operational configuration and conduct an operational checkout of the test item. Proceed with Step 5.

Before, during (between steps 4 and 5), and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, 4.5.2, 4.5.3, and 4.5.5 and shall meet the requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4, and 3.7.5 (see 3.8.1.1).

4.6.1.2 High temperature. The power supply shall be tested in accordance with Method 501, Procedure I of MIL-STD-810. Constant temperature storage shall apply (Step 3). The storage temperature shall be +160°F (+71°C). Upon completion of Step 3, adjust the internal chamber temperature to +150°F (66°C) and maintain until temperature stabilization of the test item is reached. Configure the power supply in its operational configuration and conduct an operational checkout of the test item. Proceed with Step 4.

Before, during (between steps 3 and 4), and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, 4.5.2, 4.5.3, and 4.5.5 and shall meet the requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4, and 3.7.5 (see 3.8.1.2).

4.6.2 Humidity. The Power Supply shall be subjected to the aggravated cycle humidity test specified in Method 507, Procedure 2 of MIL-STD-810. During (end of the fifth and tenth cycles) and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, 3.7.4 (see 3.8.2).

4.6.3 Low pressure (altitude). The power supply shall be tested in accordance with Method 500, Procedure I and Procedure II, of MIL-STD-810. Procedure I shall be at a pressure corresponding to 40,000 feet and Procedure II shall be at a pressure corresponding to 15,000 feet, with a rate of change not to exceed 32.8 ft/s (10 m/s).

Before, during when applicable (Procedure II, Step 3), and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, and 3.7.4 (see 3.8.3).

4.6.4 Immersion. The power supply, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810. Before and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, 3.7.4, and 3.8.4.

4.6.5 Vibration. The power supply shall be tested in accordance with Method 514, Procedure I, Category 4 (Restrained Cargo) of MIL-STD-810 using the figure for Composite two-wheeled trailer vibration exposure of Annex A. The power supply shall be vibrated not less than ½ hour on each axis. Before and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, 3.7.4, and 3.8.5.

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4.6.6 Bench handling. The power supply shall be tested in accordance with Method 516, Procedure III, of MIL-STD-810. Before and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, 3.7.4, and 3.8.5.

4.6.7 Ballistic shock. The power supply shall be tested in accordance with Method 522, Procedure VI, of MIL-STD-810. The power supply shall be secured in its normal operating position to the test plate. The test shall consist of a total of 3 blows: one each 5 foot blow, on the back, side and top of the test plate with the blows being directed to the power supply respectively. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the power supply under test is permissible. Before and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, 3.7.4, and 3.8.5.

4.6.8 Fungus. The power supply, with the cover removed, shall be subjected to the fungus test specified in Method 508 of MIL-STD-810. There shall be abundant growth colonization (see 6.5.3) on 50 percent or more of the area of the control item after 14 and 28 days. The test shall be conducted for a minimum of 28 days. No cleaning of the power supply is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the power supply. After the fungus test, the Power Supply shall be visually examined, using a 10 power magnifier. Inability of the power supply to meet the requirements of 3.8.6 shall constitute failure of this test.

4.6.9 Sand and dust. The power supply shall be tested while operating under full load in accordance with Method 510, Procedure I (Blowing Dust) and Procedure II (Blowing Sand), of MIL-STD-810. High temperature shall be high operational temperature per 3.8.1. Before and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, 3.7.4 (see 3.8.7).

4.6.10 Rain. The power supply shall be tested while operating under full load in accordance Method 506, Procedure I of MIL-STD-810. Minimum rainfall rate shall be four (4) in/hr, and minimum wind velocity shall be 40 mph Before and after testing an operational checkout shall be performed and shall consist of the tests in 4.5.1, and 4.5.2 and shall meet the requirements of 3.7.1, 3.7.3, 3.7.4 (see 3.8.8).

4.6.11 Air-seal test. The power supply shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the power supply) applied to the interior of the case. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.1 pound can be determined readily (see 3.10).

4.7 Visual and mechanical inspection. The power supply shall be inspected for the defects listed in paragraphs 3.12 and 3.12.1.

4.7.1 System safety engineering. Visual and mechanical inspection shall be performed to verify the requirements of paragraphs 3.16.1.

4.8 Electromagnetic interference (EMI). The power supply shall be tested for compliance with the requirements of 3.8 of this specification. During susceptibility tests, the power supply shall be monitored per 4.5.2 and meet the requirements of 3.7.4.

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4.9 System safety.

4.9.1 Touch current (leakage current). Equipment Touch Current shall not exceed 3.5 mA when tested IAW UL 60950-1, section 5.1 (see 3.16.1)

4.9.2 Electric strength. There shall be no insulation breakdown when a test voltage of 1500V is applied IAW UL 60950-1, section 5.2 (see 3.16.1)

4.9.3 Retained charge. The power supply shall not present a shock hazard at any external plug or point of disconnection due to stored capacitor charge IAW UL 60950-1, sections 2.1.1.7 and 2.1.1.8 (see 3.16.1).

4.9.4 Grounding continuity. Continuity shall be determined between the grounding pin of the attachment plug and accessible dead-metal parts of the power supply using a suitable continuity-indicating device such as an ohmmeter.

4.10 Reliability test. The following test shall be performed. Upon completion, test items shall meet the requirements of 3.11.

4.10.1 Early failure period. Upon completion of the early failure period test, test items shall meet the requirements of 3.11.

a) Place the test items in a test chamber capable of maintaining 149 ± 2°F (65 ± 1.1°C) and providing a single-frequency sine-wave vibration at 2.2 gravitational acceleration units peak at a non-resonant frequency between 20 Hz and 60 Hz. Connect the test items for 230 VAC, 60 Hz input and full-load output, with sensors for input and output voltage and current.

b) Set the chamber to 149 ± 2°F (65 ± 1.1°C), once the test temperature is reached, allow the test items to stabilize for not less than four hours

c) Subject power supplies to the vibration described by step a above for not less than 20 minutes without applying any load.

d) Once the vibration period is complete, subject each power supply to a load of not less than 25 amperes at 28.0 volts output with an input of 230, +0, -5 VAC and a frequency of 60 ± 5 Hz for not less than four hours.

e) Once the full-load period is complete, allow each power supply to return to room ambient conditions and stabilize for not less than two hours.

f) Conduct the tests of 4.5.1.2 and 4.7.

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4.10.2 Random failure period.

a) Place the test items in a test chamber capable of maintaining 149 ± 2°F (65 ± 1.1°C), and connect the test items for the variety of input and output conditions specified in step c) below, with sensors for input and output voltage and current.

b) Set the chamber to 149 ± 2°F (65 ± 1.1°C), once the test temperature is reached, allow the test items to stabilize for not less than four hours.

c) Load the test items in a continual sequence as follows at the same test conditions for a total of not less than four hours:

Input Voltage (AC)	Input Frequency	Load	Time
103.5, +5, -0 volts	47.5, +5, -0 Hz	No less than 25 amps	75 minutes
126.5, +0, -5 volts	63, +0, -5 Hz	No less than 20 amps	30 minutes
207, +5, -0 volts	47.5, +5, -0 Hz	No less than 25 amps	75 minutes
253, +0, -5 volts	63, +0, -5 Hz	No less than 20 amps	30 minutes
115, ± 2.5 volts	420, +0, -5 Hz	No less than 25 amps	30 minutes

d) While at the same test conditions, allow the test items to rest at no-load conditions for not less than two hours.

e) Repeat steps c) and d) continuously until a total test time of not less than 883.4 hours is obtained (441.7 hours per each of the 2 units on test).

f) Once the required test hours have been obtained, allow test items to return to room ambient conditions and stabilize for not less than two hours.

g) Conduct the tests of 4.5.1.2 and 4.7.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use The power supply is a universal tactical power supply for providing 24 to 32VDC from a nominal 115/230 VAC at 50, 60 or 400 Hz single phase. It is commonly used to replace vehicular power when using vehicular communications equipment in a tactical operations center (TOC).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a) Title, number, and date of the specification.
b) Packaging requirements (See 5.1).
c) First Article inspection
d) Marking and shipping of samples.
e) Place of final inspection.

6.3 Group C failures. Approval to ship may be withheld, at the discretion of the Government, pending decision from the contracting officer on the adequacy of corrective action (see 4.4.3.2).

6.4 Location of air-seal test. It is desirable that the air-seal test (4.6.11) be performed at a location that will minimize handling (which might cause damage to the power supply) after this inspection is completed. Use of common hardware and couplings is highly desired to facilitate Depot repair. Any preparation for shipment which would require breaking of the power supply seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Definitions.

6.5.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hyphae).

6.5.2 Chargeable failures. Chargeable failures are those that can be attributed to the item under test. Any failures that can be attributed to test equipment, test set-up, test fixtures, or due to test interruptions from power outages are not chargeable.

6.5.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.5.4 Inspection lot. An inspection lot is any group of power supplies presented to the Government for acceptance at one time.

6.5.5 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

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6.5.6 Night Vision Imaging System (NVIS) Compatibility. NVIS use image intensifier tubes to produce an enhanced image of a scene in light conditions too low for normal navigation.

6.5.7 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.5.8 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.6 Subject term (key word) listing.

Battery Standby
Single Phase

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.8 Reliability requirement and test method. The original reliability requirement for the power supplies covered by this specification was for an upper MTBF of 5,000 hours, or stated another way, a 0.0002 constant failure rate (λ, or "lambda", the reciprocal of the upper MTBF). This was erroneously translated to "90% probability of providing 5,000 hours of failure-free operation" in 2002 in an effort to comply with then DoD policies regarding changing the way reliability requirements were stated. The proper translation should have been "90% probability of 500 hours of failure free operation", since 500 hours represented the lower MTBF for $\lambda = 0.0002$ with a 90% probability.

Also in 2002, DoD policy required reliability requirements to avoid prescribing any specific test method in favor of allowing offerors to propose a test method that would be subject to technical evaluation and ranking of various proposals for a "best value" determination. In practice, this policy worked well for developing systems and major end items, but provided no benefit in re-procurement of sustainment items such as the PP-6224()/U and PP-2953()/U power supplies.

The test method described in this specification is based on test method models from the now-cancelled MIL-STD-781D and the current Revision A of MIL-HDBK-781, using an accelerated aging calculation based on the Arrhenius Acceleration Model (see section 8.7.5 of MIL-HDBK-338 for further information). In brief, testing at 65°C provides a 2.83 acceleration factor for devices intended to operate at 50°C (3.12), reducing the total hours on test from 5,000 to 1,766.8 hours. By increasing the duty cycle from four hours per 12-hour period to four hours for each 6-hour period, another acceleration factor of 2 was obtained, reducing the total hours to 883.4 hours.

Custodian:
Army – CR

Preparing activity:
Army – CR
(Project 6130-2010-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.daps.dla.mil/online/.

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